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Spacer

Implementation Document

Smart Short-Let Management & Fair Distribution System

Prepared for Court 474 Ltd

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1. PROJECT OVERVIEW

Court 474 Ltd is developing a short-let apartment complex consisting of **25 one-bedroom apartments** and **4 two-bedroom apartments**, along with a **40-seater conference room**. These apartments will be constructed and sold to multiple individual owners (subscribers). Court 474 will then manage the short-let bookings, guest access, and revenue distribution on behalf of these owners.

The central challenge is **fairness**. With 29 apartments owned by different people, the system must ensure that bookings and revenue are distributed equitably across all units. No owner should feel that their apartment is being overlooked while others receive preferential treatment.

This document outlines how **Spacer** — Algolog's property management platform already live and processing bookings across multiple properties in Nigeria — will be deployed for Court 474, with its **Fair Distribution Algorithm** configured specifically for the multi-owner model.

What Spacer delivers for Court 474:

- A branded booking link where guests browse, select dates, and pay online
- A fair distribution algorithm that automatically assigns apartments to bookings
- TTLock smart lock integration — access codes tied to the guest's period of stay
- Revenue splitting — agency charge to Court 474, remainder to the apartment owner
- Owner dashboard with full transparency into bookings, earnings, and fairness metrics
- Cart abandonment recovery — automatic reminders until the booking is completed
- Temporary space removal — owners can block their apartment from bookings
- Conference room booking as a separate revenue stream
- Smart home synchronisation advisory — TTLocks, smart lighting, and automated shutoff

2. THE PROBLEMS WE ARE SOLVING

A. Fair Guest Distribution Across Multiple Owners

With 29 apartments owned by different subscribers, manual allocation of guests is inherently unfair. Without a system, some apartments will receive more bookings than others — whether by coincidence, convenience, or unconscious bias. Owners will inevitably question whether they are getting their fair share, and this erodes trust in Court 474's management.

The solution requires an algorithm that distributes bookings across all apartments in a way that is transparent, auditable, and mathematically fair — while still being practical (considering availability, apartment type, and owner preferences).

B. Revenue Splitting and Transparency

Every booking generates revenue that must be split between Court 474 (agency/management fee) and the apartment owner. This needs to happen automatically with a clear audit trail. Owners need to see exactly how much their apartment earned, what fee was deducted, and what their net payout is — without having to ask or chase.

C. Incomplete Bookings

Prospective guests who start a booking but don't complete payment represent lost revenue. Without follow-up, these bookings are simply abandoned. The system needs to automatically remind these guests until they either complete the booking or the hold period expires.

D. Owner Control Over Their Space

Apartment owners may need to temporarily remove their unit from the booking pool — for personal use, maintenance, renovation, or any other reason. The system must allow owners to block specific dates without affecting other apartments or the overall booking flow.

E. Smart Access & Key Management

Physical key handovers are impractical at scale with 29 apartments. Guests need self-service access that activates on arrival and deactivates on checkout — without a front desk or manual coordination.

F. Conference Room as a Separate Revenue Stream

The 40-seater conference room needs its own booking system (hourly and daily), separate from the apartment bookings, with its own availability calendar and payment processing.

3. THE FAIR DISTRIBUTION ALGORITHM

This is the core intelligence of the Spacer deployment for Court 474. The algorithm ensures that every apartment gets a fair share of bookings and revenue over time, while still making practical allocation decisions based on real-time conditions.

How It Works

When a guest initiates a booking (e.g., "1-bedroom apartment, April 10–15"), the algorithm evaluates all eligible apartments and selects one based on the following criteria, applied in order:

Step 1: Filter by Type

If the guest requests a 1-bedroom, only the 25 one-bedroom apartments are considered. If they request a 2-bedroom, only the 4 two-bedroom apartments are considered. This is a hard filter — no cross-type allocation.

Step 2: Filter by Availability

Remove any apartments that are already booked for the requested dates, or that have been temporarily blocked by the owner. Only apartments that are genuinely available proceed to the next step.

Step 3: Filter by Readiness

Remove any apartments that are not in a "ready" state — e.g., under maintenance, or flagged for any operational reason. Only guest-ready apartments proceed.

Step 4: Rank by Fairness Score

Each apartment maintains a **fairness score** that the algorithm updates after every booking. The score considers two factors:

- **Booking count** — How many bookings has this apartment received in the current period (month/quarter)? Apartments with fewer bookings score higher.
- **Revenue earned** — How much total revenue has this apartment generated? This matters because a 2-bedroom apartment earning ₦200,000 from 2 bookings shouldn't be treated the same as a 1-bedroom earning ₦60,000 from 2 bookings. The algorithm balances by revenue, not just count.

The apartment with the **lowest fairness score** (fewest bookings + lowest revenue in the current period) gets priority for the next booking.

Step 5: Randomised Tie-Breaking

If multiple apartments have the same fairness score (common at the start of a new period), the algorithm selects randomly among them. This prevents any predictable pattern that could feel unfair.

Distribution Modes

Court 474 management can choose between three distribution modes depending on their preference:

Mode	How It Works	Best For

Weighted Fair Share	Balances by both booking count and revenue earned. Default mode.	Multi-owner properties where fairness is the top priority
Round-Robin	Strict rotation: Apt 1 → Apt 2 → Apt 3… regardless of revenue	When owners want perfectly equal booking counts
Manual Override	Admin assigns a specific apartment. Logged in audit trail.	Special requests, VIP guests, or operational reasons

Fairness Transparency

Every owner can see their fairness metrics from their dashboard:

- Total bookings received vs. property average
- Total revenue earned vs. property average
- Distribution history — which apartment was selected for each booking and why
- Period-over-period fairness trend (are things balancing out over time?)

This transparency is what builds trust. Owners don't have to take Court 474's word for it — they can verify fairness themselves at any time.

4. THE BOOKING FLOW

This is the end-to-end journey from a guest discovering Court 474 to checking out of their apartment.

Step 1: Guest Visits Booking Link

Court 474 gets a branded booking page (e.g., court474.spacer.so or a custom domain). The guest browses available apartment types, views photos and descriptions, and selects their preferred dates.

Step 2: Algorithm Selects Apartment

The guest does not choose a specific apartment — they choose a type (1-bed or 2-bed) and dates. The fair distribution algorithm automatically selects the most appropriate apartment based on the fairness score, availability, and readiness criteria described above.

Step 3: Payment

The guest pays online via Paystack (card or bank transfer). Payment is confirmed instantly. The booking is locked and the apartment is marked as occupied for those dates.

Step 4: TTLock Access Code Generation

A unique TTLock smart lock access code is generated and sent to the guest via email and SMS. This code is tied to the guest's period of stay — it activates on check-in day and automatically expires at checkout. No physical key handover required.

Step 5: Guest Stay

The guest arrives, uses their TTLock access code, and stays in the apartment. If they need to extend, they can do so through the platform (subject to availability). Court 474 staff can monitor all active stays from the admin dashboard.

Step 6: Checkout

On the checkout date, the TTLock access code expires automatically. The apartment is marked as available and re-enters the booking pool for the next guest.

Step 7: Revenue Settlement

After each booking, the system automatically calculates the revenue split: Court 474's agency charge is deducted, and the remaining amount is credited to the apartment owner's account. Settlement reports are generated automatically.

5. CART ABANDONMENT RECOVERY

When a prospective guest starts a booking but does not complete payment, the system does not simply discard the booking. Instead, it enters the **cart abandonment recovery flow**:

1. **Immediate hold** — The selected apartment is temporarily held for a defined period (e.g., 30 minutes) to give the guest time to complete payment.
2. **First reminder** — If payment is not completed within the hold period, the guest receives an email and/or SMS reminder with a direct link to complete their booking.
3. **Follow-up reminders** — The system sends follow-up reminders at configured intervals (e.g., after 2 hours, after 24 hours, after 48 hours) until either the guest completes the booking or the maximum reminder count is reached.
4. **Release** — If the guest does not complete payment after all reminders, the apartment hold is released and the unit returns to the available pool for other guests.

This recovery flow is critical because every abandoned booking is potential revenue. In our experience with other Spacer deployments, cart abandonment recovery can recapture 15–25% of bookings that would otherwise be lost.

6. TEMPORARY SPACE REMOVAL

Apartment owners may need to temporarily remove their unit from the booking pool. The platform provides this through a **calendar blocking** feature:

- **Owner-initiated blocking** — The owner logs into their dashboard and blocks specific dates. The apartment is removed from the booking pool for those dates. No bookings can be assigned to it during the blocked period.
- **Reason tracking** — The system records the reason for the block (personal use, maintenance, renovation, etc.) for management visibility.
- **Advance notice** — Blocking can only be done for future dates. Dates with existing confirmed bookings cannot be blocked (the owner would need to coordinate with Court 474 management to relocate an existing booking).
- **Fairness adjustment** — When an apartment is blocked by the owner, the fairness algorithm accounts for this. The blocked period is excluded from the fairness calculation so the owner is not penalised for voluntarily removing their unit.
- **Auto-return** — Once the blocked period ends, the apartment automatically returns to the booking pool. No manual action needed.

7. REVENUE SPLITTING & OWNER SETTLEMENTS

Every booking generates revenue that is split between Court 474 and the apartment owner. The platform handles this automatically:

The Revenue Flow

1. Guest pays for booking (e.g., ₦50,000 for 3 nights)
2. Payment lands in Court 474's Paystack account
3. System calculates the split:

• **Court 474 agency charge** — a percentage of the booking amount (e.g., 20%)

• **Owner net amount** — the remainder (e.g., 80%)

4. The split is recorded and visible to both Court 474 management and the apartment owner
5. Court 474 disburses to the owner on an agreed schedule (weekly or monthly)

Example:

Item	Amount
Guest pays (3 nights × ₦15,000/night)	₦45,000
Paystack processing fee (1.5%)	– ₦675
Net received	**₦44,325**
Court 474 agency charge (20%)	₦8,865
Owner net payout	**₦35,460**

Note: The agency charge percentage is configurable and agreed between Court 474 and its owners. It can be set globally or per apartment if different owners have different agreements.

Owner Settlement Reports

Each owner receives automated settlement reports showing:

• Each booking on their apartment (dates, guest, amount)

• Gross amount received

• Agency charge deducted

• Net payout amount

• Running total for the period

• Comparison to property average (fairness metric)

8. CONFERENCE ROOM BOOKING

The 40-seater conference room operates as a separate revenue stream with its own booking flow:

- **Hourly and daily booking options** — clients can book by the hour or for a full day
- **Online availability calendar** — visible on the booking page, shows available slots in real time
- **Online payment** — same Paystack integration, instant confirmation
- **Add-ons** — projector, catering, and other services can be added as optional extras
- **Access instructions** — sent automatically upon booking confirmation
- **Separate marketing** — the conference room can be marketed independently from the apartments

Conference room revenue goes entirely to Court 474 (unless a different arrangement is agreed with apartment owners).

9. SPACER FEATURES FOR COURT 474

All of the following are existing Spacer capabilities that will be deployed for Court 474:

Feature	Description
Branded Booking Portal	Custom booking page with property photos, apartment types, date selection, and online payment
Fair Distribution Algorithm	Apartment selection ensuring equitable bookings and revenue across all owners
TTLock Smart Lock Integration	Unique access codes generated per booking, tied to stay period, auto-expire at checkout
Cart Abandonment Recovery	Automatic reminders to guests who start but don't complete bookings
Owner Dashboard	Each owner sees bookings, revenue, fairness metrics, and can block dates for personal use
Revenue Splitting	Automatic agency charge deduction and owner settlement with full audit trail
Temporary Space Removal	Owners block dates for personal use/maintenance; apartment auto-returns when period ends
Conference Room Booking	Separate hourly/daily booking flow with availability calendar and payment
Admin Dashboard	Management sees all bookings, occupancy, revenue, settlements, and fairness metrics
Automated Notifications	Booking confirmations, access codes, reminders, settlement reports via email and SMS
Real-Time Occupancy	Visual grid showing occupied, available, and blocked apartments at a glance

Invoice & Receipt Generation	Automatic for all bookings — exportable as PDF
Conflict Prevention	Prevents double-bookings across the entire property

10. SMART ACCESS & SMART HOME ADVISORY

TTLock Smart Door Locks

We recommend **TTLock** smart locks for all 29 apartments and the conference room. TTLock is the smart lock brand that Spacer integrates with natively. These locks support:

- **Temporary passcodes** — generated by Spacer per booking, valid only for the stay period
- **Auto-expiry** — codes deactivate automatically at checkout time
- **Remote management** — lock/unlock remotely from the admin dashboard
- **Audit trail** — every lock/unlock event is logged with timestamp
- **Battery powered** — no wiring required, easy to install on existing doors
- **Bluetooth + Wi-Fi gateway** — TTLock G2 gateway enables remote access over the internet

Procurement: Court 474 procures the TTLock hardware directly. Algolog handles the software integration — connecting each lock to Spacer so access codes are generated and managed automatically with every booking.

Fully Smart Apartment System

Beyond door locks, Court 474 has expressed interest in a fully smart apartment setup where everything can be controlled and automated. We recommend the following smart home components:

Smart Lighting

Wi-Fi or Zigbee smart bulbs/switches that can be scheduled, dimmed, or turned off remotely. Can be configured to turn off automatically when the guest's access code expires (checkout).

Smart Switches & Sockets

Replace standard wall switches with smart switches. Allows remote control of all electrical outlets — lights, AC, water heater. Everything can be shut off centrally when the apartment is unoccupied.

Smart AC Control

IR blasters or smart thermostats that control air conditioning units remotely. AC can be pre-cooled before guest arrival and shut off on checkout.

Automated Shutoff on Departure

When a guest's TTLock code expires or they check out, the system can trigger a "shutoff all" command — turning off lights, AC, water heater, and non-essential sockets. This reduces electricity waste between guests.

Energy Monitoring

Smart meters or energy monitoring plugs to track electricity consumption per apartment. Useful for billing owners or identifying wasteful usage patterns.

Procurement: Court 474 procures all smart home hardware (switches, bulbs, IR blasters, energy monitors) directly from suppliers. Algolog provides the advisory on which specific products are compatible and handles the software synchronisation — connecting the smart home devices to

Spacer so they respond to booking events (check-in, checkout, code expiry).

11. INVESTMENT

A. Spacer Platform Setup & Deployment

This covers the full Spacer deployment for Court 474 — booking portal, fair distribution algorithm, payment processing, revenue splitting, owner dashboards, cart abandonment recovery, conference room booking, admin dashboard, TTLock integration, and all automated notifications.

Item	Cost (NGN)
Spacer Platform Setup & Deployment (29 apartments + conference room)	Included
Fair Distribution Algorithm Configuration	Included
TTLock Software Integration (up to 30 locks)	Included
Revenue Splitting Configuration	Included
Cart Abandonment & Notification System	Included
Training & Handover	Included
Total Spacer Deployment	**₦10,000,000**

B. Smart Home Synchronisation Setup

This covers the software integration of smart home devices (lighting, switches, AC control, automated shutoff) with the Spacer platform — so they respond to booking events automatically. Hardware procurement is handled separately by Court 474.

Item	Cost (NGN)
Smart Home Device Integration with Spacer (up to 29 apartments)	Included
Automated Shutoff Logic (checkout triggers all-off)	Included
Energy Monitoring Dashboard Integration	Included
Device Advisory & Compatibility Testing	Included
Total Smart Home Synchronisation	**₦2,000,000**

Component	Cost (NGN)
Spacer Platform Setup & Deployment	₦10,000,000
Smart Home Synchronisation Setup	₦2,000,000
Total Investment	**₦12,000,000**

Note: The above covers all software setup, configuration, integration, and training. Hardware procurement (TTLock door locks, smart switches, smart bulbs, IR blasters, energy monitors, Wi-Fi gateways) is handled directly by Court 474. Algolog will provide a detailed hardware specification list with recommended models and quantities.

WHY ALGOLOG

Algolog Limited is a Nigerian technology company committed to ensuring businesses across Africa get the technology they need to build their future, improve their sales, and strengthen their operations.

Our work includes:

- **Spacer** (spacer.so) — Smart hospitality and workspace management platform, deployed across multiple properties in Abuja, Kaduna, and Lagos
- **DeepThread** (deepthread.ai) — AI-powered cross-platform context engine
- **Visora** (visora-ai.com) — AI-powered data analytics and insights platform
- **PrivateTransfer** (privatetransfer.ng) — Secure transfer platform

Spacer is already live and processing bookings at properties across Nigeria. Every feature described in this document is a production-ready Spacer capability. Court 474 is getting a proven, deployed platform — not a promise.

NEXT STEP

We'd love to continue the conversation — the owner agreements, the agency charge structure, the smart home hardware specifications, and the deployment timeline.

Whenever you're ready, reach out to:

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